

Exploring the Potential to Reduce Aviation's Non-CO2 Climate Impact through Controlled Plume Intersections

Kieran Tait



A dissertation submitted to the University of Bristol in accordance
with the requirements for award of the degree of
PhD in Aerospace Engineering in the Faculty of Engineering.

School of Civil, Aerospace and Design Engineering

November 2024

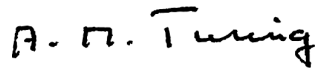
Word count: TBD

Abstract

Acknowledgements

Author's Declaration

I declare that the work in this dissertation was carried out in accordance with the requirements of the University's Regulations and Code of Practice for Taught Programmes/Research Degree Programmes (delete as appropriate) and that it has not been submitted for any other academic award. Except where indicated by specific reference in the text, the work is the candidate's own work. Work done in collaboration with, or with the assistance of, others, is indicated as such. Any views expressed in the dissertation are those of the author.

A handwritten signature in black ink that reads "A. N. Turing". The signature is written in a cursive style with a large, stylized 'T'.

Kieran Tait

2 November 2024

Contents

List of Tables	iv
List of Figures	v
1 Introduction	1
1.1 Motivation	1
1.1.1 Aircraft Emissions	2
1.1.2 Radiative Forcing of Aircraft Emissions	2
1.1.3 Aircraft Exhaust Plume Treatment	3
1.1.4 Plume Superposition in High-Density Airspace	4
1.1.5 Atmospheric Response to Plume Superposition	4
1.1.6 Mitigation Potential	6
1.2 Thesis Objectives	7
1.3 Thesis Structure	8
2 Potential to Offset Aviation NO_x-induced Warming Through Formation Flight and Chemical Saturation	10
2.1 Introduction	11
2.2 Methodology	15
2.2.1 Determination of Threshold NO _x from Kinetic Data	15

Contents

2.2.2	Determination of Threshold NO_x from STOCHEM-CRI CTM Modeling Data and IAGOS Measurement Data	19
2.2.3	Development of 1-D Box Model of Flight Corridor	20
2.3	Results and Discussion	21
2.3.1	$[\text{NO}_2]_{\text{equiv}}$ in the Troposphere Determined from Kinetic Data . . .	21
2.3.2	Global Mapping of $[\text{NO}_2]_{\text{equiv}}$ at the Cruise Level	22
2.3.3	Aircraft Plume and its Chemistry Study using a 1D Dispersion Model	25
2.4	Conclusion	33
3	Development of a Novel Aircraft Plume Modelling Framework	35
3.1	Introduction	35
3.2	GPAT Overview	36
3.3	Input Parameters	39
3.3.1	Flight Parameters (fl_params)	39
3.3.2	Plume Parameters (plume_params)	40
3.3.3	Simulation Parameters (sim_params)	41
3.4	Datasets	42
3.4.1	Flight Dataset (FL)	43
3.4.2	Meteorology Dataset (MET)	44
3.4.3	Background Chemistry Dataset (BG_CHEM)	46
3.4.4	Plume Dataset (PL)	46
3.4.5	Emissions Dataset (EMI)	46
3.4.6	Contrail Dataset (CONTRAIL)	46
3.4.7	Chemistry Dataset (CHEM)	46
3.5	Models and Methods	46
3.5.1	Trajectory Model (traj_gen)	46

Contents

3.5.2	Meteorology Model (gen_met)	46
3.5.3	Background Chemistry Model (gen_bg_chem)	46
3.5.4	Aircraft Performance Model (ac_perf)	46
3.5.5	Emissions Model (emissions)	46
3.5.6	Dispersion/Advection Model (sim_plumes)	46
3.5.7	Plume-to-Grid Model (plume_to_grid)	46
3.5.8	Contrail Model (run_cc)	46
3.5.9	Chemistry Model (run_boxm)	46
3.5.10	Validation	46
3.6	Conclusion	46
4	Plume Overlap as A Climate Solution – A Sensitivity Study	47
5	Conclusion	48

List of Tables

1	Introduction	1
2	Potential to Offset Aviation NO_x-induced Warming Through Formation Flight and Chemical Saturation	10
3	Development of a Novel Aircraft Plume Modelling Framework	35
4	Plume Overlap as A Climate Solution – A Sensitivity Study	47
5	Conclusion	48

List of Figures

1	Introduction	1
2	Potential to Offset Aviation NO_x-induced Warming Through Formation Flight and Chemical Saturation	10
2.1	Calculated [NO ₂] _{equiv} using simple kinetic equation (Equation (2.2)) at different temperatures and altitudes.	18
2.2	Global [NO ₂] _{equiv} at model output level 8 (9.2–11.8 km) estimated using [CH ₄], [CO], [M], and temperature fields simulated by STOCHEM-Common Representative Intermediate (CRI).	23
2.3	Global [NO ₂] _{equiv} at model output level 8 (9.2–11.8 km) estimated using [CO], [M], and temperature fields from In-service Aircraft for Global Observing System (IAGOS) dataset and [CH ₄] from STOCHEM-Common Representative Intermediate (CRI) model simulation.	24
2.4	NO _x mixing ratios as a function of distance from the flight path at increasing times after the passage of an aircraft, each offset by 3 ppb. . .	27
2.5	Aircraft plume width as a function of travel time.	28
2.6	O ₃ mixing ratios as a function of distance from the flight path at increasing times after the passage of an aircraft, each offset by 3 ppb.	30

List of Figures

2.7	Time development of the average O ₃ mixing ratios across the 99.9 km grid for a base case with one aircraft, five aircraft flying together and flying with horizontal separations increasing from 100 m to 10 km.	32
3	Development of a Novel Aircraft Plume Modelling Framework	35
3.1	GPAT process flow schematic portraying the input parameters (rhombi), constituent models (boxes) and flow of data through a range of datasets (cylinders).	38
4	Plume Overlap as A Climate Solution – A Sensitivity Study	47
5	Conclusion	48

1 Introduction

Aircraft act as high-altitude emissions vectors, transporting a number of radiatively and chemically active substances across vast regions of the globe. These substances induce a net global warming effect that constitutes 3.5% of the global climate impact due to anthropogenic sources [1]. Two-thirds of this impact result from aircraft non-CO₂ emissions, which have a variable impact depending on when and where they are released. This means that avoiding flying in climate-sensitive regions of the atmosphere has the potential to reduce aviation's climate impact significantly [2]. Furthermore, the environmental conditions within the aircraft exhaust plume result in additional chemical and microphysical processing that is not accounted for in global chemistry models. In high-density airspace, such as along the North Atlantic Flight Corridor, aircraft often fly in close proximity, giving rise to the superposition of their plumes and the accumulation of emissions contained within them [3]. It has been shown in [4] that the overlap of four consecutive plumes can influence the nonlinear chemistry and microphysics significantly, leading to saturation effects that can potentially provide a net climate impact reduction.

1.1 Motivation

It is the aim of this research to simulate this plume overlap concept over a range of atmospheric conditions and aircraft frequencies, to investigate the sensitivity of the climate response. The simulation results can then be used to test the mitigation potential

of plume superposition through the application of aircraft trajectory optimisation and formation flight.

1.1.1 Aircraft Emissions

For aircraft to achieve and maintain steady level flight, a propulsion system is often required, providing the necessary thrust force to counteract aerodynamic drag and generate sufficient lift to overcome the aircraft's weight. In the commercial aviation industry, which is responsible for 88% of global aviation fuel usage [Gossling2020], it is commonplace to use a gas turbine propulsion system operating on kerosene-based jet fuel. The stoichiometric combustion of kerosene with air produces the mechanical energy required to operate the engine and generate thrust, however it also emits a number of additional chemical species into the aircraft wake in the process. The emission species which are of great chemical and radiative importance are carbon dioxide (CO_2), carbon monoxide (CO), water vapour (H_2O), nitrogen oxides ($\text{NO}_x = \text{NO} + \text{NO}_2$), sulphur dioxide (SO_2) and particulate matter (PM).

1.1.2 Radiative Forcing of Aircraft Emissions

The most commonly used metric to compare the magnitudes of climate impact from a range of emission species is radiative forcing (RF). RF is defined as the perturbation to the net energy balance of the Earth-atmosphere system due to natural or anthropogenic factors of climate change, measured in watts per square metre [W/m^2].

The RF of aviation CO_2 emissions is direct and well understood. Due to its thermodynamic and photochemical stability, CO_2 has a relatively long atmospheric lifetime on the order of 100 to 1,000 years. This means that aviation CO_2 emitted into the atmosphere simply serves to increase its atmospheric concentration, leading to the eventual distribution over global spatial scales and a steady, uniform increase in the planetary greenhouse

1 Introduction

effect. The non-CO₂ aircraft emission species however, are much more chemically active and hence have much shorter lifetimes and a more localised impact. They primarily contribute to climate change through two indirect radiative forcing mechanisms. Firstly, chemically active species such as NO_x and CO react with the surrounding air to modify chemical concentrations of key climate agents such as ozone (O₃) and methane (CH₄). Secondly, emissions of H₂O and PM are responsible for the formation of condensation trails (contrails) and may even trigger the generation of additional cirrus clouds, both of which can affect the Earth's radiative balance. With the chemical and meteorological state of the atmosphere being highly variable with respect to both location and time, this implies that aircraft non-CO₂ emissions have a variable climate impact depending on when and where they are released. The spatio-temporal sensitivity of the climate to non-CO₂ aircraft emissions becomes particularly important when observing the chemical and physical effects that occur in the aircraft exhaust plume.

1.1.3 Aircraft Exhaust Plume Treatment

Following expulsion into the free atmosphere, exhaust species become entrained in the aircraft wake, forming a plume of elevated chemical concentrations which persist for 2 to 12 hours [EPA1992], before fully dispersing into ambient air. The build-up of non-CO₂ emissions in the plume gives rise to a number of nonlinear chemical and microphysical effects, which influence the ensuing atmospheric response by altering the net production rates of radiatively active gases and affecting contrail formation and persistence. It is often the case however, that in large-scale atmospheric models, emissions are instantaneously diluted into the volume of the smallest resolved grid cell, with dimensions according to the model's spatial resolution. The instantaneous dilution (ID) approach neglects the plume-scale processes, thus leading to discrepancies in the calculated climate response such as the overestimation of O₃ production, CH₄ and CO destruction, and the increased

1 Introduction

rate of NO_x conversion to nitrogen reservoir species [6]. Furthermore, it is stated in [7] that the formation of ice in aircraft exhaust plumes may result in additional heterogeneous chemical reactions, that are not captured in global atmospheric models.

1.1.4 Plume Superposition in High-Density Airspace

Air traffic levels correlate with aviation demand, giving rise to a spatial and temporal inhomogeneity in the distribution of aircraft, and hence their emissions, across the globe. In a spatial sense, demand tends to be highest around and between densely-populated regions such as the United States, Europe and Asia, meaning flights often take place over mainland airspace and along the flight corridors connecting such regions. In a temporal sense, demand varies both diurnally (time of day) and seasonally (time of year), favouring daylight hours and the summer months [8].

Airspace regions that are particularly susceptible to high demand may see aircraft fly at minimum separation distances from one another. This close proximity flying leads to the superposition of aircraft exhaust plumes, and the accumulation of emissions entrained within them. [3] carries out an in-situ investigation into air traffic emissions signatures in the North Atlantic Flight Corridor (NAFC), finding that the superposition of 2-5 plumes lead to the levels of NO_x , SO_2 and PM exceeding background concentrations by factors of 30, 5 and 3, respectively. This highlights the severity with which amassing aircraft emission species can alter the local chemical composition of the atmosphere. Such changes to atmospheric conditions can significantly affect the non- CO_2 climate response of a flight through further chemical and microphysical processing.

1.1.5 Atmospheric Response to Plume Superposition

In [4], the notion of plume superposition and the extent to which it influences the atmospheric response was explored, by comparing results from an expanding plume

1 Introduction

model against an atmospheric model which assumes instant dilution. The analysis was carried out firstly on a single aircraft at cruise altitude, then on four aircraft flying in the same direction each separated by 1 hr. Observing the single aircraft's plume 10 hrs after emission, it was found that the instantaneous dilution approach overestimated O₃ production by 33%, CH₄ destruction by 30% and CO destruction by 32%. When observing the response due to the four overlapping plumes however, the overestimation of O₃ production increases to 77%, CH₄ destruction to 68% and CO destruction to 74%.

The marked difference in measured response for four overlapping plumes compared to a single plume serves as evidence that emissions saturation in superimposed plumes can significantly alter the chemical response of the atmosphere. However, further work needs to be done to clarify the direction and magnitude of the net change in climate impact resulting from these changes to the chemical composition. Moreover, the sensitivity of the chemical and climate response to plume superposition needs to be tested for a range of aircraft transit frequencies and ambient atmospheric conditions, so as to build up a visual representation over the phase space of the atmosphere.

In addition to chemical changes in the atmosphere, plume overlap can also influence the formation and persistence of aircraft contrails and the subsequent generation of aviation-induced cirrus clouds. [9] explored the effect of recurrent water vapour emission on contrail formation and persistence at typical aircraft cruising altitudes. It was discovered that continual contrail generation actually leads to dehydration of the surrounding atmosphere and a diminishing radiative forcing effect of ~15% in the affected areas. This outcome provides further impetus to explore the potential effects of plume superposition on the global warming induced by aviation.

1.1.6 Mitigation Potential

The significance of the plume overlap effect and its potential for influencing the atmospheric response brings forth the potential for controlled implementation of this effect, as a means of aviation climate impact mitigation. A near-term fuel burn reduction measure that is currently popular in the literature is the concept of formation flight, which has the potential to provide a complementary non-CO₂ climate reduction by exploiting the benefits of plume superposition. Formation flight involves the flight of two or more aircraft in aerodynamic formation, with the follower aircraft positioned in the smooth updraft of the leader aircraft's wake, reducing required lift and thrust, hence reducing fuel and CO₂ emissions by 5-8% per trip [10]. A fortuitous outcome of formation flight is however, the saturation of emissions in the trailing exhaust plumes, which can lead to the aforementioned nonlinear chemical and microphysical response that reduces ozone and contrail formation.

The non-CO₂ climate effects of a twin-aircraft formation flight are observed in [11], where a climate model was used to determine the changes in NO_x-related ozone production and contrail formation processes. The case study found that, despite a 1-3% increase in flown distance, CO₂ was reduced by 6% and NO_x was reduced by 11%. This resulted in a 5% reduction in ozone production efficiency due to NO_x saturation and a 48% contrail reduction due to mutual competition for available water vapour, resulting in a total climate impact reduction of approximately 23%. While part of the alleviated climate impact can be related to the decrease in total emissions due to wake energy retrieval of the follower aircraft, there is also further emphasis due to the saturation of emissions in the overlapping plumes of both aircraft involved. This demonstrates great potential for reducing both CO₂- and non-CO₂-induced climate effects from aviation, if such a scheme were to be carried out on a global scale.

Fleet-wide implementation of formation flight would however, pose a number of

research obstacles, that must be overcome to maximise its cost-benefit potential, whilst ensuring the safe and orderly operation of aircraft in controlled airspace. Optimising aircraft trajectories to minimise fuel burn requires the consideration of nonlinear aircraft performance, wind and weather, payload, fuel load, and constraints set out by air traffic control [12]. Formation flight adds another variable to the fuel burn optimisation problem, maximising the time spent flying in formation whilst minimising deviation from the true optimal route. This is a research topic covered extensively in the literature [13; 14]. Optimising formation flight trajectories to minimise climate impact on the other hand, is a relatively novel concept, which requires deeper understanding of the sensitivity of the climate response to emissions under a range of different atmospheric states.

1.2 Thesis Objectives

The studies performed in [4] and [9] provide a thought-provoking insight into the notion of aircraft plume superposition, and [11] demonstrates its potential utilisation for aviation climate impact mitigation through formation flight. This thesis aims to drastically expand the scope of analysis conducted into chemical saturation effects that occur in high-density airspace. It intends to do so by satisfying the following objectives:

- To conduct an expansive literature review into the state-of-the-art research surrounding aircraft emissions, aircraft exhaust plume dynamics, air traffic management and distribution, the local and global climate effects induced by aviation, and lastly, the mitigation of aviation climate impact, with a focus on operational strategies targeting non-CO₂ emissions.
- To utilise real-world air traffic data and first-order plume dispersion methods to assess the operational feasibility of aircraft exhaust plume overlap in today's airspace.

1 Introduction

- To investigate the global spatial and temporal variability of NO_x saturation effects, using a world-leading atmospheric chemistry transport model (CTM) STOCHEM-CRI, which incorporates the chemistry scheme CRIv2-R5 [x].
- To develop a modelling framework to robustly investigate the chemical and physical saturation effects that influence the non- CO_2 climate impact of formation flights.
- To carry out a localised study on aircraft NO_x saturation effects using this modelling framework, in which parameters such as aircraft transit frequency and air temperature will be varied.

1.3 Thesis Structure

In essence, the thesis is aiming to solve one fundamental question:

“What is the sensitivity of the climate response to the superposition of two or more aircraft exhaust plumes, under a variety of atmospheric conditions?”

As such, the document is structured as follows: Chapter 2 introduces the science behind aviation climate impact, and explores a wide range of literature pertaining to aircraft emissions, their plume-scale effects, and the spatio-temporal sensitivity of the atmospheric response. Chapter 3 investigates the feasibility of aircraft exhaust plume intersections in modern airspace, through consideration of real-world air traffic data and rudimentary dispersion calculations, to estimate degree of plume overlap in high-density regions. In chapter 4, the potential to offset climate change through formation flying is investigated in more detail, with a focus on gas-phase chemical kinetics associated with NO_x emissions and their varying impact on ozone formation. This leads to the development of a localised plume chemistry modelling framework in chapter 5. In this chapter, the methodology behind a novel aircraft plume analysis tool is outlined and supporting models and methods are introduced. The model is also validated against

1 Introduction

its predecessor. In chapter 6, this modelling framework is then utilised in a formation flight study, to investigate the most critical parameters affecting the severity of the NO_x saturation phenomenon, and the benefits and drawbacks of formation flying in the context of aviation climate impact mitigation.

2 Potential to Offset Aviation NO_x-induced Warming Through Formation Flight and Chemical Saturation

The content of this chapter is based on the following published journal article:

Khan, M. A. H.; Tait, K. N.; Derwent, R. G.; Roome, S.; Bacak, A.; Bullock, S.; Lowenberg, M. H.; Shallcross, D. E., "Off-setting climate change through formation flying of aircraft, a feasibility study reliant on high fidelity gas-phase chemical kinetic data", *International Journal of Chemical Kinetics*, vol. 55, no. 7, pp. 402–412, 2023. DOI: <https://doi.org/10.1002/kin.21644>.

This chapter incorporates material from that work with permission from the publisher and co-authors, where necessary. Dr Anwar Khan was lead author, and together with Prof. Dudley Shallcross, provided the theory and methodology for the study, and conducted the global chemistry analysis using STOCHEM-CRI. I co-wrote the paper with Dr Khan and Prof. Shallcross, carrying out further analysis and validation using the In-service Aircraft for a Global Observing System (IAGOS) dataset.

2.1 Introduction

Gas-phase kinetic studies underpin our understanding of key atmospheric processes such as stratospheric ozone formation, regulation and depletion [Rowland2006; Dameris2010], and air pollution (e.g., NO_x, tropospheric ozone, peroxy acetyl nitrate (PAN), and aerosol) generation and reduction [VonSchneidemesser2015; Stockwell2019]. Research into this field is thus crucial for air quality and climate change assessment and mitigation. The aircraft sector is an important component in the assessment of air pollution and climate change, as it is one of only two non-surface sources of NO_x emissions [Wasiuk2016; Tait2022; Schumann1997], with the other being lightning [Tie2002; Schumann2007; Stockwell1999]. In addition to the direct emission of CO₂ through jet fuel combustion, aircraft emit NO_x which contributes to tropospheric ozone formation, as well as water vapor and particulates that give rise to contrail formation (Figure 1) [Lee2021].

Traditionally, mitigation efforts have gone toward incremental increases in efficiency, due to technological advancements and innovative design. However, due to the increasing urgency to rapidly reduce aviation climate impact, the focus has shifted toward more radical measures. This includes the development of new propulsion systems (e.g., liquid hydrogen) [Yusaf2022; Khan2022], sustainable aviation fuels (SAFs) [Doliente2020], and changes to aircraft flight routes and operations. Unlike the former two methods, modifications to flight routing pose the most viable solution for implementation this decade [Niklass2019]. This paper explores the potential of formation flying to reduce the ozone production efficiency of aircraft NO_x emissions.

Reducing NO_x emissions requires a lower temperature burn and this conflicts with trends toward higher temperature, more energy efficient jet fuel combustion. Formation flight may hold the key to maximizing the reduction in ozone production, due to any additional NO_x that is output from aircraft. The concept involves two or more aircraft flying no more than 1–2 km apart longitudinally, akin to energy-saving migratory flocks

2 Potential to Offset Aviation NO_x-induced Warming Through Formation Flight and Chemical Saturation

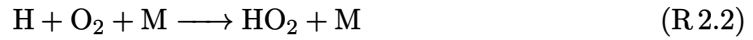
of geese. This enables follower aircraft to “surf” the wake of their leader, thus reducing required lift and hence thrust and fuel burn, on the order of 5% per aircraft [Xu2014; Bower2009].

Further to the fuel saving achieved through this concept, there are also two notable chemical saturation effects that occur, due to the local accumulation of aircraft emissions. First, in airspace conditions conducive to persistent contrail formation, formation flight can help reduce the net contrail effect. This is because the generation of two or more persistent contrails in the same region will compete for available water vapor. Thus, they mutually inhibit growth and the combined warming effect is lessened compared with if they were generated individually. Unterstrasser et al. [Unterstrasser2020] explores this effect in great detail.

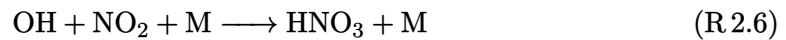
The second saturation effect to be observed is the accumulation of NO_x emissions, leading to NO_x-saturated conditions and a negative correlation with ozone production efficiency. It is well known in the urban environment, where there can be very high NO_x levels, that ozone formation is virtually zero as OH collapse occurs and NO₂ is converted to nitric acid (HNO₃) [Jenkin2020; Sillman2020]. HNO₃ eventually is incorporated into aerosol and removed from the atmosphere, representing a permanent sink for NO_x and preventing ozone formation. Can formation flying also deliver a reduction in ozone formation following NO_x emission? If NO_x levels are maintained at high enough levels, OH collapse could occur; this sounds bad, but along flight corridors, a significant fraction of NO₂ may be sequestered to form HNO₃ and be unable to contribute to ozone formation. The chemical kinetics is in the aircrafts’ favor. In essence, when emitted, NO_x can, under high enough VOC levels, catalyze the formation of ozone, for example, using CO as a representative VOC, even though it is not strictly speaking one.



2 Potential to Offset Aviation NO_x-induced Warming Through Formation Flight and Chemical Saturation



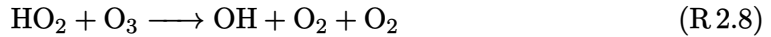
Therefore, through reactions (R 2.1)–(R 2.5), ozone is produced through NO₂ photolysis, and the longer the chain length converting NO to NO₂ (R 2.3) and back to NO (R 2.4) the more ozone is formed. However, if NO and NO₂ levels are high enough, the reaction of NO₂ with OH (reaction (R 2.6)) can start to compete with the reaction of OH with CO and CH₄ in the mid to upper troposphere (reactions (R 2.1) and (R 2.7)).



The reason why reaction (R 2.6) may start to compete in the upper troposphere, where temperatures are far lower than they are at the surface, is that as an association reaction, the rate coefficient of the reaction (R 2.6), k_6 , increases with decreasing temperature [Atkinson2004; Butkovskaya2005] whereas rate coefficient of the reaction (R 2.7), k_7 decreases with decreasing temperature [Atkinson2003] in a typical Arrhenius type behavior. Rate constant of the reaction (R 2.1), k_1 has both a direct and association channel but appears to vary very little with temperature.

2 Potential to Offset Aviation NO_x-induced Warming Through Formation Flight and Chemical Saturation

Aircraft NO_x perturbs the fast photochemistry of the upper troposphere–lower stratosphere (UTLS) by interfering with the balance between photochemical ozone production and ozone destruction. This balance is controlled by the NO_x-compensation point, the point at which with decreasing NO_x levels, photochemical ozone production through reaction (R 2.3) exactly balances photochemical ozone destruction through:



In the UTLS region, the NO_x compensation point, which is set by the rate coefficients for the reactions (R 2.3) and (R 2.8), is about 10 ppt as NO (calculated assuming the rate coefficient of reaction (R 2.3), k_3 of $1.19 \times 10^{-11} \text{cm}^3 \text{molecule}^{-1} \text{s}^{-1}$, the rate coefficient of reaction (R 2.8), k_8 of $1.13 \times 10^{-15} \text{cm}^3 \text{molecule}^{-1} \text{s}^{-1}$ based on Atkinson et al. [Atkinson2004] and O₃ 100 ppb). That is to say, if the NO mixing ratio is less than 10 ppt, the fast HO_x photochemistry leads to ozone destruction. This is the general state of the UTLS region under typical conditions of temperature and pressure. In the presence of aircraft at cruise altitude, through their NO_x emissions, the NO_x compensation point may be exceeded and photochemical ozone production sets in, leading ultimately to an increase in global warming since the cold conditions of the UTLS favor enhanced radiative forcing from the increased ozone mixing ratios. The elevated OH concentrations driven by reaction (R 2.3) lead to elevated CH₄ destruction and hence decreased build-up, with its attendant negative radiative forcing and global cooling.

Most of the studies of the impacts of aircraft cruise emissions have been completed using sophisticated three-dimensional (3-D) Eulerian grid-based chemistry-transport models (CTMs). Despite a high degree of sophistication, typical spatial resolutions are no better than $1^\circ \times 1^\circ$ (say, 100 km $\times$ 100 km in the North Atlantic air traffic corridor). A review of these studies is available elsewhere [Fuglestedt1999]. Current CTM studies of these greenhouse gas impacts may well overstate the impacts of aircraft

cruise emissions. This arises because of the current mis-match between the coarse spatial resolution of the CTMs and the fine-scale dimensions of the major air traffic corridors. A similar conclusion may be reached by consideration of the temporal resolution of these studies. The CTM studies may necessarily average aircraft cruise emissions over months or even years whereas on average flights last hours [Wasiuk2016]. Together, the lack of spatial and temporal CTM resolution may ensure that aviation NO_x is present in excess of the NO_x compensation point at all times and in all grid squares through which a significant number of aircraft fly.

In this paper, first a kinetic model feasibility study was carried out to determine what NO_x threshold level is required to start reducing ozone formation. Second, an established 3D global CTM, STOCHEM-CRI, was used to identify regions of the atmosphere where the lowest NO_x thresholds exist. Thirdly, validation of the results from the global analysis of NO_x threshold concentrations was conducted by running the simulations with a much higher resolution dataset, and comparing these results with that derived from coarse STOCHEM-CRI model. The In-service Aircraft for Global Observing System (IAGOS) data (<https://www.iagos.org/iagos-data/>) is used in place of the 5 °× 5 °monthly global model fields from STOCHEM-CRI [Khan2021]. Finally, we develop a simple aircraft plume dispersion model which addresses the issues of inadequate spatial and temporal CTM resolution and understands how different the chemistry is likely to look under conditions of higher spatial and temporal resolution.

2.2 Methodology

2.2.1 Determination of Threshold NO_x from Kinetic Data

An estimate of the threshold level of NO₂ defining [NO₂]_{equiv} as the concentration of [NO₂] required to start OH collapse can be made by comparing the loss rates for OH with CH₄ (reaction (R 2.6)) and CO (reaction (R 2.1)) with that for OH with NO₂ (reaction

2 Potential to Offset Aviation NO_x-induced Warming Through Formation Flight and Chemical Saturation

(R 2.6)).

$$k_6[\text{NO}_2]_{\text{equiv}}[\text{OH}][\text{M}] = k_1[\text{CO}][\text{OH}] + k_7[\text{CH}_4][\text{OH}] \quad (2.1)$$

A simple rearrangement of (2.1) provides an expression for $[\text{NO}_2]_{\text{equiv}}$:

$$\begin{aligned} [\text{NO}_2]_{\text{equiv}} &= \frac{k_1[\text{CO}][\text{OH}] + k_7[\text{CH}_4][\text{OH}]}{k_6[\text{OH}][\text{M}]} \\ &= \frac{k_1[\text{CO}] + k_7[\text{CH}_4]}{k_6[\text{M}]} \end{aligned} \quad (2.2)$$

The temperature dependent rate coefficient of the reaction (R 2.1), $k_1 = 5.4 \times 10^{-14} \times ((T/298)^{1.5}) \times \exp(250/T)$ and reaction (R 2.7), $k_7 = 2.45 \times 10^{-12} \times \exp((-1775 \pm 100)/T)$ have been obtained from Baulch et al. [Baulch1992] and DeMore et al. [DeMore1997], respectively. The reaction (R 2.6) depends not only on temperature, but also on pressure. A decrease in pressure reduces the value of the rate coefficient and so there is competition between these two components with altitude: increasing altitude leads to a decrease in temperature and an increase in k_6 , whereas pressure decreases, and this will decrease k_6 . The term, $k_6[\text{M}]$ was calculated as:

$$k_6[\text{M}] = \frac{k_0[\text{M}]}{1 + \frac{k_0[\text{M}]}{k_\alpha}} \times 0.6 \left(\frac{1}{1 + \left(\log \frac{k_0[\text{M}]}{k_\alpha} \right)^2} \right) \quad (2.3)$$

where $k_0 = (2.5 \pm 0.1) \times 10^{-30} (T/300)^{-4.4 \pm 0.3}$ and $k_\alpha = (1.6 \pm 0.2) \times 10^{-11} (T/300)^{-1.7 \pm 0.2}$, taken from DeMore et al. [DeMore1997].

Temperature and pressure decrease with increasing altitude in the troposphere for a given latitude and longitude, but for a given altitude, temperature decreases from the equator to the poles. Although CH₄ has a hemispheric gradient, higher in the Northern Hemisphere (~1800 ppbv) than the Southern Hemisphere (~1700 ppbv), and a slow

2 Potential to Offset Aviation NO_x-induced Warming Through Formation Flight and Chemical Saturation

inter-hemisphere transport with a temporal scale of approximately 1 year, mixing ratios of CH₄ are pretty constant throughout the troposphere as a result of its 10 year lifetime [Shallcross2020]. Whereas, CO, whose lifetime is around a month, has a strong gradient between Northern Hemisphere (120–180 ppbv) and Southern Hemisphere (60–70 ppbv), and displays mixing ratios that drop off considerably with altitude [Shallcross2020]. If we imagine, for simplicity, that CO is 1100 ppb at the surface and 100 ppb at 10 km and from surface to 10 km drops off by 100 ppb km⁻¹, we can start to make some preliminary calculations to scope out where in the atmosphere recreating the polluted urban atmosphere may be most likely.

2 Potential to Offset Aviation NO_x-induced Warming Through Formation Flight and Chemical Saturation

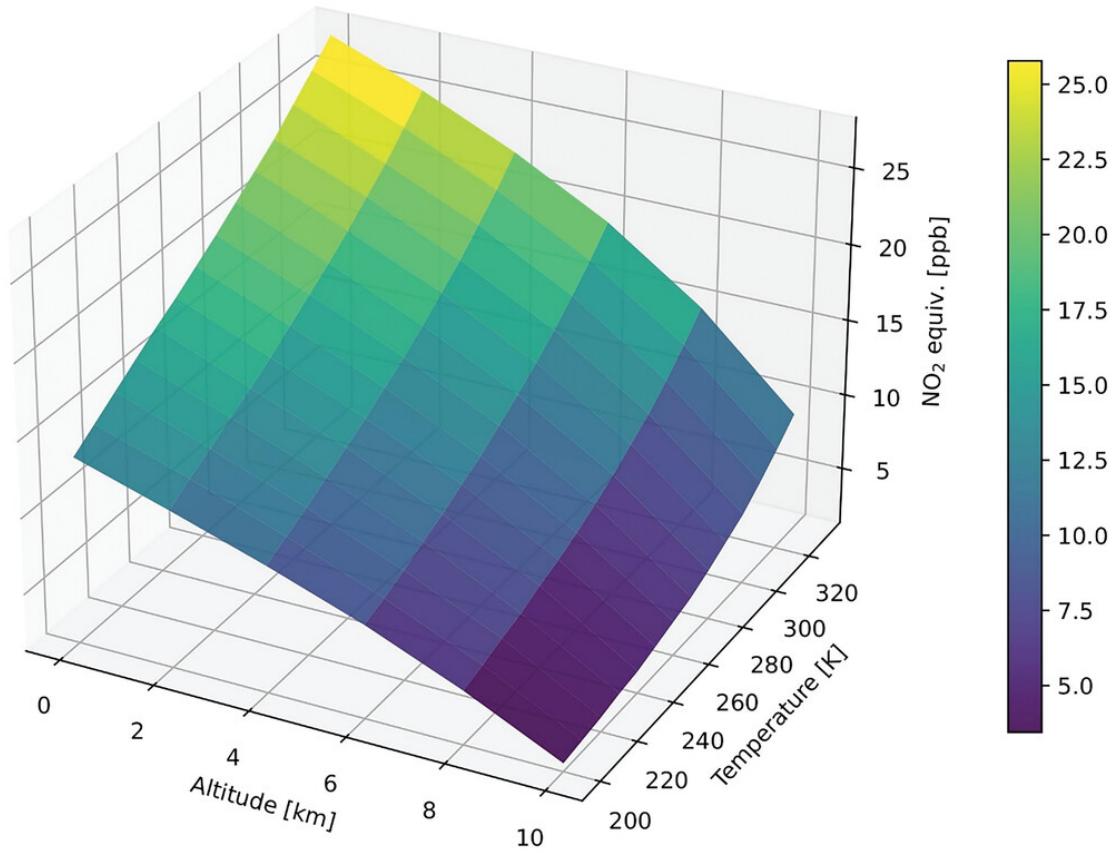


Figure 2.1: Calculated $[\text{NO}_2]_{\text{equiv}}$ using simple kinetic equation (Equation (2.2)) at different temperatures and altitudes.

Therefore, the $[\text{NO}_2]_{\text{equiv}}$ was calculated as a function of temperature typical of the troposphere (200–330 K) and at pressures from the surface up to 10 km using the simple CO mixing ratio altitude profile noted earlier, with a fixed mixing ratio of CH₄, 1900 ppb.

2.2.2 Determination of Threshold NO_x from STOCHEM-CRI CTM Modeling Data and IAGOS Measurement Data

The same calculation as shown in section ?? was repeated using global model concentrations of CH₄, CO, and the rate coefficients as a function of temperature and pressure. These model fields were computed at a resolution of 5 °longitude × 5 °latitude with altitude binning corresponding to model output levels from Collins et al. [Collins1997]. The monthly global model fields of temperature, OH, CO, and CH₄ were obtained from the output of a 3-D chemistry and transport model, STOCHEM-CRI [Fuglestedt1999]. These global fields provided a basis for calculation of [NO₂]_{equiv} on a global scale to examine the spatial and temporal variations of this parameter. To further confirm the NO_x threshold trends seen in the global model analysis, the calculations were performed again, this time on measurement data from the three IAGOS projects: MOZAIC, CORE, and CARIBIC [Blot2021]. In these datasets, datapoints have been generated from measurements of key climate variables, using scientific instruments onboard civil aircraft. The MOZAIC project ran from 1994 to 2010 and collected data from over 33,471 flights. CORE collected data from 2011 up until present day, providing data on a further 7790 flights. Finally, CARIBIC has been operating since 1997 to present day, however, has only collected data from 541 flights. These datasets were used to create temperature, M and CO concentration fields, which are comparable to the corresponding STOCHEM-CRI model fields. These fields were constructed by collating data in latitude, longitude, altitude bins, which match the resolution of the STOCHEM-CRI simulated fields. These fields are used in the calculation of NO_x threshold which are compared with that found from STOCHEM-CRI.

Measurements are taken at a nominal temporal resolution of 4 s throughout the entire flight profile. Anomalous or missing data led to sparsity between measurements; however, data quality has generally improved in recent years, as newer and more accurate equipment

2 Potential to Offset Aviation NO_x-induced Warming Through Formation Flight and Chemical Saturation

was deployed. To perform these calculations, all datapoints were removed from flights with either no useful compositional chemical data or where vital physical measurements such as temperature, pressure, or altitude were missing. These steps reduced our sample size down to 16,335 individual flights with 83.8 million useful datapoints, collected over a 28-year period.

The necessary computations were carried out using the C API to netCDF and Slurm workload manager on the BlueCrystal Phase 4 supercomputing facility (<https://www.acrc.bris.ac.uk/acrc/phase4>). Outputs were stored in CSV format and queried using an SQL database, where we binned data into the grid cell resolution of STOCHEM-CRI. Then averages of our estimated $[\text{NO}_2]_{\text{equiv}}$ values were calculated for each box in our world grid. The global distribution plots were generated using Python packages: Matplotlib and Cartopy.

2.2.3 Development of 1-D Box Model of Flight Corridor

A simple one-dimensional (1-D) model of aircraft cruise conditions was developed which addresses the issues of inadequate spatial and temporal CTM resolution. A set of grid boxes has been built perpendicular to an air traffic corridor in the UTLS region. The grid contains 999 boxes, each of $100 \text{ m} \times 100 \text{ m} \times 100 \text{ m}$ and the contents are assumed to be instantaneously mixed. Each grid box contains a number of trace gases, that is, representing the chemistry of the UTLS region including CH_4 , CO , NO_x , O_3 , H_2O , organic compounds, and their degradation products. The time-dependent behavior of each trace gas is represented by a differential equation of the form:

$$\frac{\partial c_i}{\partial t} = P_i - l_i \cdot c_i - \frac{\partial}{\partial x} K \frac{\partial c_i}{\partial x} \quad (2.4)$$

where c_i is the mass of species, i , t the time, x the axis perpendicular to the air traffic corridor, P_i the production rate of the species within the box, and l_i is the rate coefficient for the loss of the species within the box and the final term on the right-

hand side of the equation represents the gain and loss of the species by exchange with neighboring boxes with the grid. Equation (2.4) was expressed in finite-difference form and integrated numerically with a dispersion time-step of 300 s and a chemical time-step of 20 s using a backwards-iterative Euler numerical method. K is the horizontal diffusion coefficient and was set at $10 \text{ m}^2\text{s}^{-1}$ based on Paoli et al. [Paoli2011] and Schumann et al. [Schumann1995] for the UTLS region. The chemical production and loss rates were taken from the Common Representative Intermediate (CRI) mechanism [Jenkin2008; Watson2008; Utembe2009] which describes the chemistry of 220 species taking part in 609 reactions. Background conditions for the UTLS were taken from a previous run of STOCHEM-CRI for 40° N , June 21 at 150 mb and used to provide initial conditions for each of the 999 grid boxes.

2.3 Results and Discussion

2.3.1 $[\text{NO}_2]_{\text{equiv}}$ in the Troposphere Determined from Kinetic Data

Figure 2.1 illustrates that the $[\text{NO}_2]_{\text{equiv}}$ determined from kinetic data decreases with temperature and altitude and that the most likely place where the equivalent point may be reached is in fact around the cruise altitude of long-haul aircraft. Short-haul aircraft may not be able to attain such altitudes if the flight is very short in duration. The simple model presented in figure 2.1 suggests that as the flight reaches cruise altitude at about 10 km, the amount of NO_2 required for parity in OH loss is around 2 ppb. How feasible is it to generate such levels of NO_2 ? In a narrow flight corridor with frequent flights through it, it may well be.

2.3.2 Global Mapping of $[\text{NO}_2]_{\text{equiv}}$ at the Cruise Level

Figure 2.2 shows global $[\text{NO}_2]_{\text{equiv}}$ at model output level 8 which is determined by using $[\text{CH}_4]$, $[\text{CO}]$, $[\text{M}]$, and temperature fields simulated by STOCHEM-CRI. In this study, data at model output level 8 (9.2–11.8 km) were used, to best replicate the conditions experienced by civil aircraft in cruise. It is known that CO mixing ratios are very variable in space and time, for example, in the biomass burning season in the tropics, CO in the upper troposphere can be elevated considerably, which could be responsible for the spatial and temporal variations of global $[\text{NO}_2]_{\text{equiv}}$ level.

Figure 2.2 suggests that sub-ppb levels of $[\text{NO}_2]_{\text{equiv}}$ may exist in mid- and high-latitudes, but that several ppb (up to 2 ppb) are required in the tropics, where temperature is higher and CO levels elevated through deep convection of CO from the surface sources (e.g., vegetation, biomass burning, anthropogenic emissions, VOC oxidation). The biomass burning season for Northern Hemisphere Africa and South East Asia is from November to March with peak burning in December and January; however, the biomass burning seasons for South America and Southern Africa show a peak in September to October with little burning between November and February, which is responsible for the monthly variations of $[\text{NO}_2]_{\text{equiv}}$ peaks at these areas (see Figure 2.2). There is little biomass burning over central Europe and North America, but the strong anthropogenic source along with VOC oxidation leads to a higher level of CO in these areas, resulting in ~1 ppb $[\text{NO}_2]_{\text{equiv}}$. The vegetation emission of CO can also increase $[\text{NO}_2]_{\text{equiv}}$ at Amazon and South Africa forest regions during Austral summer months (December–January–February).

To further confirm the NO_x threshold, termed $[\text{NO}_2]_{\text{equiv}}$, trends seen in the global model analysis in Figure 2.2, the results using IAGOS data are plotted in Figure 2.3. The $[\text{NO}_2]_{\text{equiv}}$ values obtained from both STOCHEM and IAGOS show a good agreement (see Figures 2.2 and 2.3) with the differences between all months averaged $[\text{NO}_2]_{\text{equiv}}$ at level 8 referred to by mean bias (STOCHEM-IAGOS) and the percentage change

2 Potential to Offset Aviation NO_x-induced Warming Through Formation Flight and Chemical Saturation

((STOCHEM-IAGOS)*100/IAGOS) are found to be 0.147 ppb and 0.27%, respectively. The calculated global $[\text{NO}_2]_{\text{equiv}}$ from both model and measurement data (Figures 2.1 and 2.2) suggested that the northern midlatitudes of the atmosphere are the most favorable region existing with the smallest NO_x thresholds (0.5 ppb) needed before reduction in ozone formation is likely to occur at cruise altitude of aircraft.

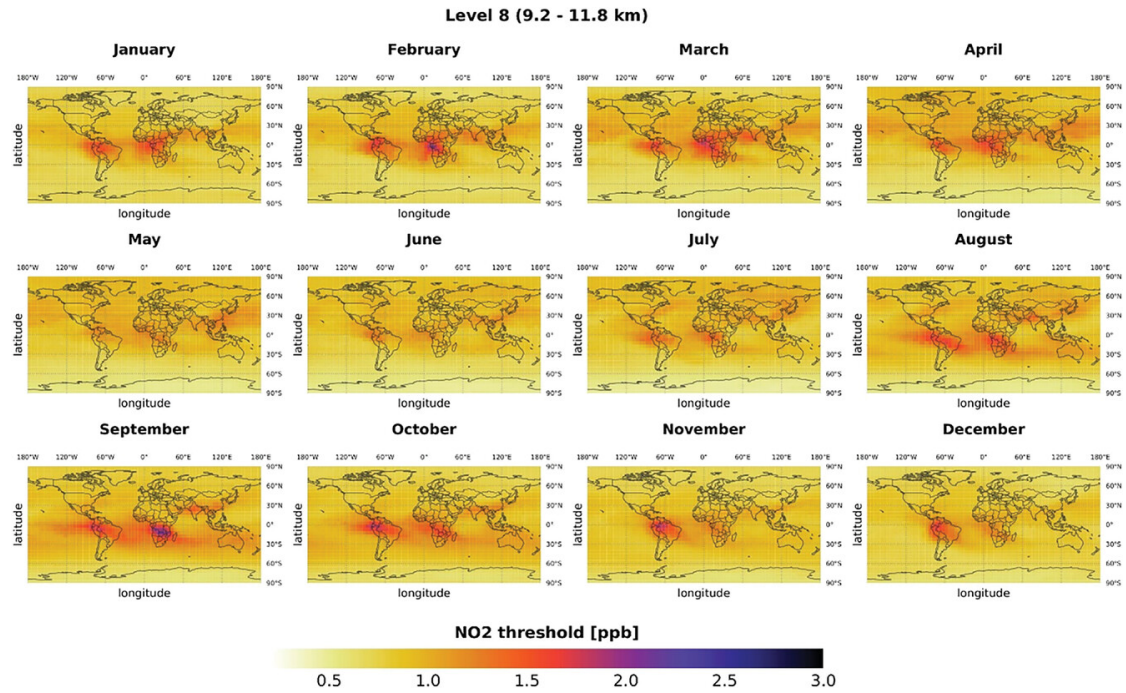


Figure 2.2: Global $[\text{NO}_2]_{\text{equiv}}$ at model output level 8 (9.2–11.8 km) estimated using $[\text{CH}_4]$, $[\text{CO}]$, $[\text{M}]$, and temperature fields simulated by STOCHEM-Common Representative Intermediate (CRI).

2 Potential to Offset Aviation NO_x-induced Warming Through Formation Flight and Chemical Saturation

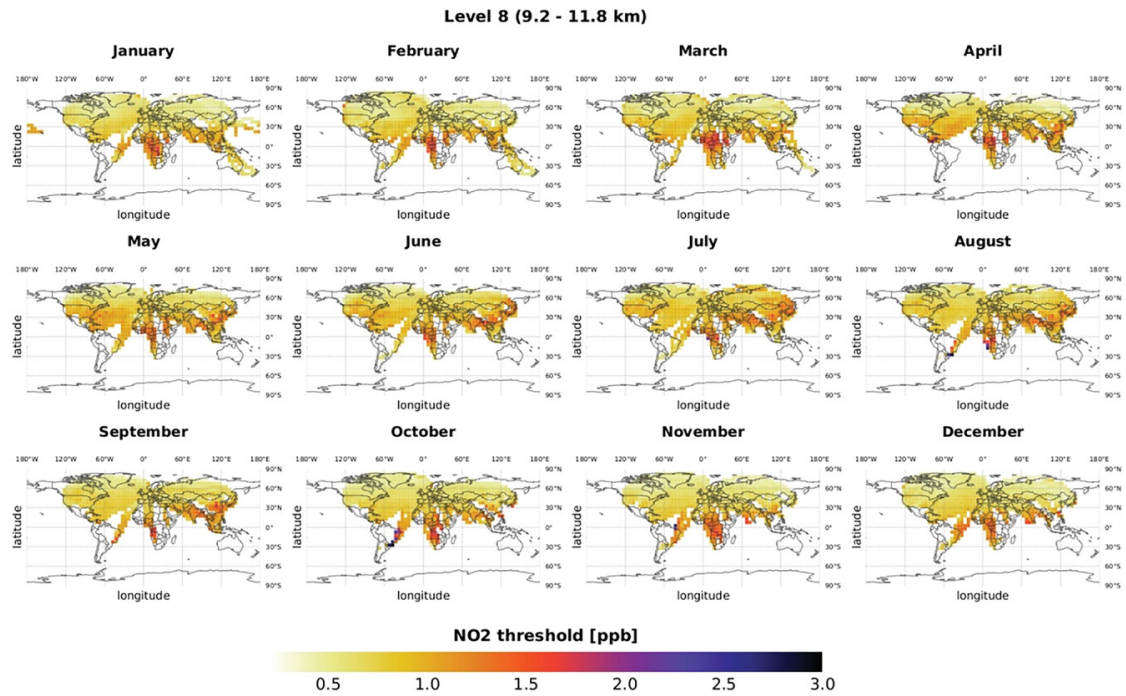


Figure 2.3: Global $[\text{NO}_2]_{\text{equiv}}$ at model output level 8 (9.2–11.8 km) estimated using $[\text{CO}]$, $[\text{M}]$, and temperature fields from In-service Aircraft for Global Observing System (IAGOS) dataset and $[\text{CH}_4]$ from STOCHEM-Common Representative Intermediate (CRI) model simulation.

Assessing how well NO_x is sequestered by OH radicals to form HNO₃, terminating ozone production, will depend on k_6 (function of pressure and temperature) and this itself will change with altitude, latitude, and time of year. As altitude increases for a given location, temperature decreases and so k_6 will increase. However, at the same time pressure is decreasing and so k_6 will decrease, determining which process wins and whether k_6 ultimately decreases or increases with altitude requires high fidelity kinetic information. An increasing k_6 with altitude may mean that this termination reaction is more effective at high altitudes for a given $[\text{NO}_2]$. These reactions are hard to study

2 Potential to Offset Aviation NO_x-induced Warming Through Formation Flight and Chemical Saturation

at low temperature and as a function of pressure and so uncertainties increase. The rate coefficients (k_1 , k_6 , and k_7) were perturbed upwards and downwards by considering the uncertainties of the rate coefficients and used to calculate global annual average $[\text{NO}_2]_{\text{equiv}}$ (see Supplementary information Figure ??). The uncertainty analysis shows that $[\text{NO}_2]_{\text{equiv}}$ can be decreased and increased by up to 2.5% and 5% in the northern midlatitudes due to the using lower limit and higher limit rate coefficients, respectively (Figure ??).

The calculation of $[\text{NO}_2]_{\text{equiv}}$ did not consider VOCs contributions in the study. The methane mixing ratios have been rising in the troposphere since ca. 2010 but most VOCs are decreasing. For ethane and propane [Pozzer2022], two of the longest lived VOCs, and species that will have elevated concentrations in the upper troposphere, current measured concentrations are indeed decreasing [Pozzer2022], and suggest that these VOCs would contribute less than 5% of the OH loss compared with CH₄. Given that loss of OH with CH₄ is typically 30% of that for CO, VOCs other than CO and CH₄ can be ignored in any initial analysis such as carried out here. Even for acetone, another long lived species with elevated levels in the upper troposphere higher levels are seen in the summer months (800 ppt) but drop to around 200 ppt in winter [Dufour2016]. So, for short periods of time in a year, levels approach that required to be around 5% of that for CH₄ but for the Northern Hemisphere, where it is expected the formation flying to be most effective, levels are much lower.

2.3.3 Aircraft Plume and its Chemistry Study using a 1D Dispersion Model

In a first box model experiment, an aircraft emits a pulse of NO_x into the center box of the model grid at 12:00 z. The pulse size was set based on a global aircraft cruise NO_x emission of 3.9 Tg NO_x per year and a distance traveled [5] of 36.4 billion km that is

2 Potential to Offset Aviation NO_x-induced Warming Through Formation Flight and Chemical Saturation

107 g NO_x km⁻¹. This NO_x was instantaneously mixed into the center box volume and the time development of the model species was then followed. After several seconds, the NO-NO₂-O₃ chemistry was rapidly established, a strong depletion in ozone formed and then NO_x began to disperse horizontally into the neighboring boxes over the period of hours and days. After 1 h the horizontal distribution of NO_x showed an aircraft plume with a width of 3 km, as shown in Figure 2.4, which expanded to 6.8 km after 6 h and 13.1 km after 24 h. Plume width expanded with the square root of lapsed time, see Figure 2.5, exactly as predicted by the Einstein–Smoluchowski equation.

2 Potential to Offset Aviation NO_x-induced Warming Through Formation Flight and Chemical Saturation

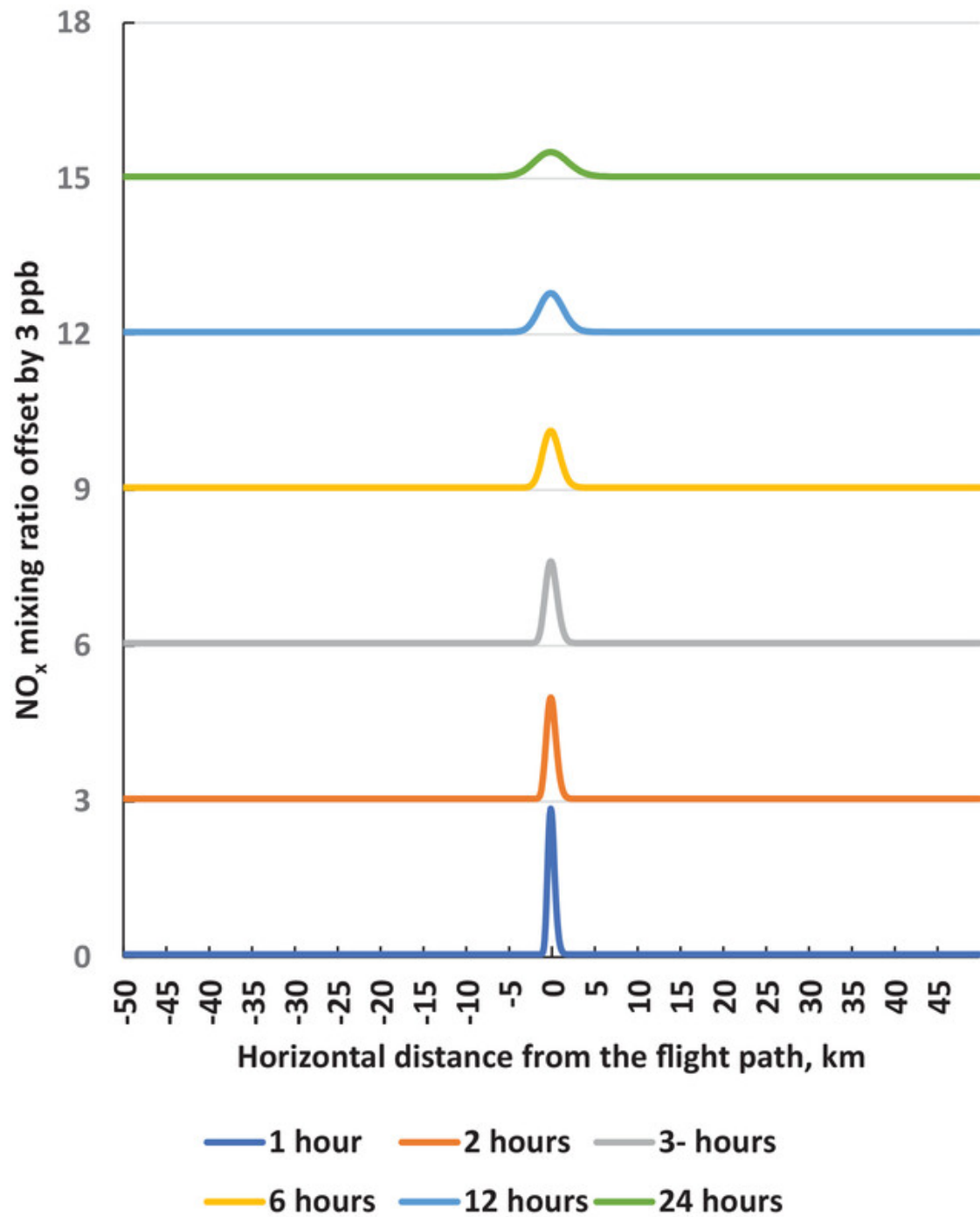


Figure 2.4: NO_x mixing ratios as a function of distance from the flight path at increasing times after the passage of an aircraft, each offset by 3 ppb.

2 Potential to Offset Aviation NO_x-induced Warming Through Formation Flight and Chemical Saturation

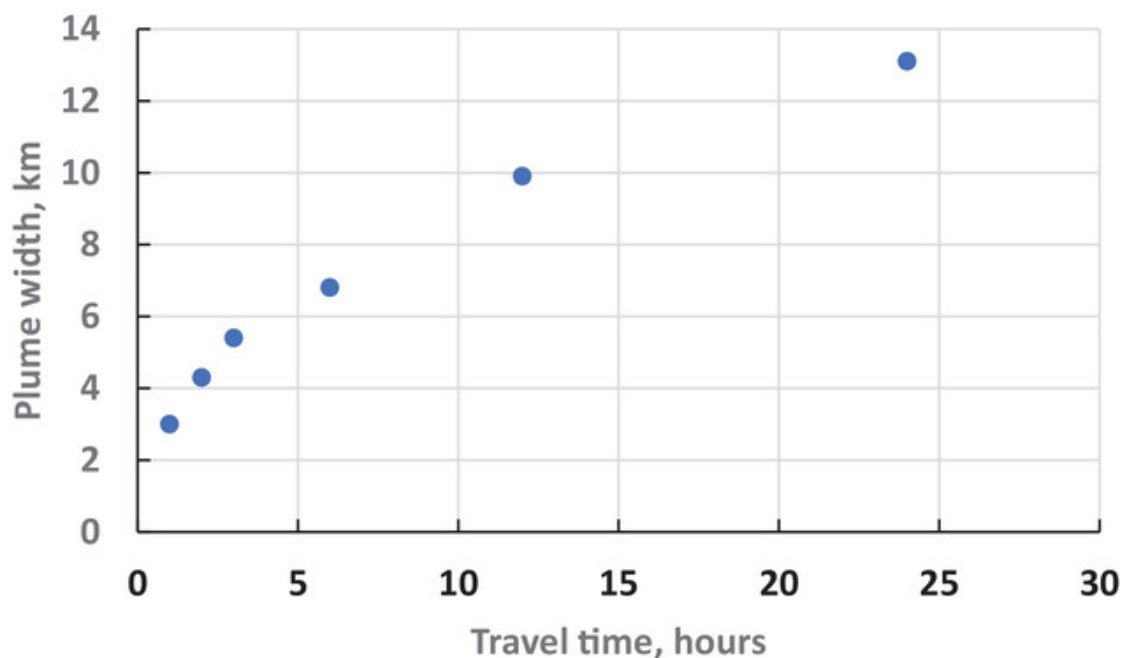


Figure 2.5: Aircraft plume width as a function of travel time.

Initially all of the ozone would have been completely wiped out in the center of the flight-path by its rapid reaction with NO:



However, during daytime, the product NO₂ is rapidly photolyzed to form ozone. Overall, the reaction sequence (R 2.3)–(R 2.5) rapidly returns the mixture of NO–NO₂–O₃ to a steady state in a few minutes in which the ozone levels are largely restored to their values before the passage of the aircraft.

The distribution of ozone perpendicular to the flight path is shown in Figure 2.6 for various times after the passage of the aircraft. One hour after the aircraft passage, ozone levels show the presence of a slight dip close to the center of the aircraft plume demonstrating that the stationary state in the NO–NO₂–O₃ system has been re-established.

2 Potential to Offset Aviation NO_x-induced Warming Through Formation Flight and Chemical Saturation

After 3 h, the central dip is still evident but photochemical ozone production is starting to appear in the wings of the aircraft plume. As time passes, the production in the wings becomes more evident and the central dip is filled in. After 24 h, photochemical ozone production is well on its way and an ozone plume becomes apparent with a width of about 16 km and a height of just over 0.4 ppb at its center. At this stage, the ozone plume is about 3 km wider than the NO_x plume as horizontal dispersion moves the photochemically produced ozone into the background atmosphere. After 4 days, see Figure 2.6, the plume is about 30 km wide and has a height of about 2.7 ppb.

2 Potential to Offset Aviation NO_x-induced Warming Through Formation Flight and Chemical Saturation

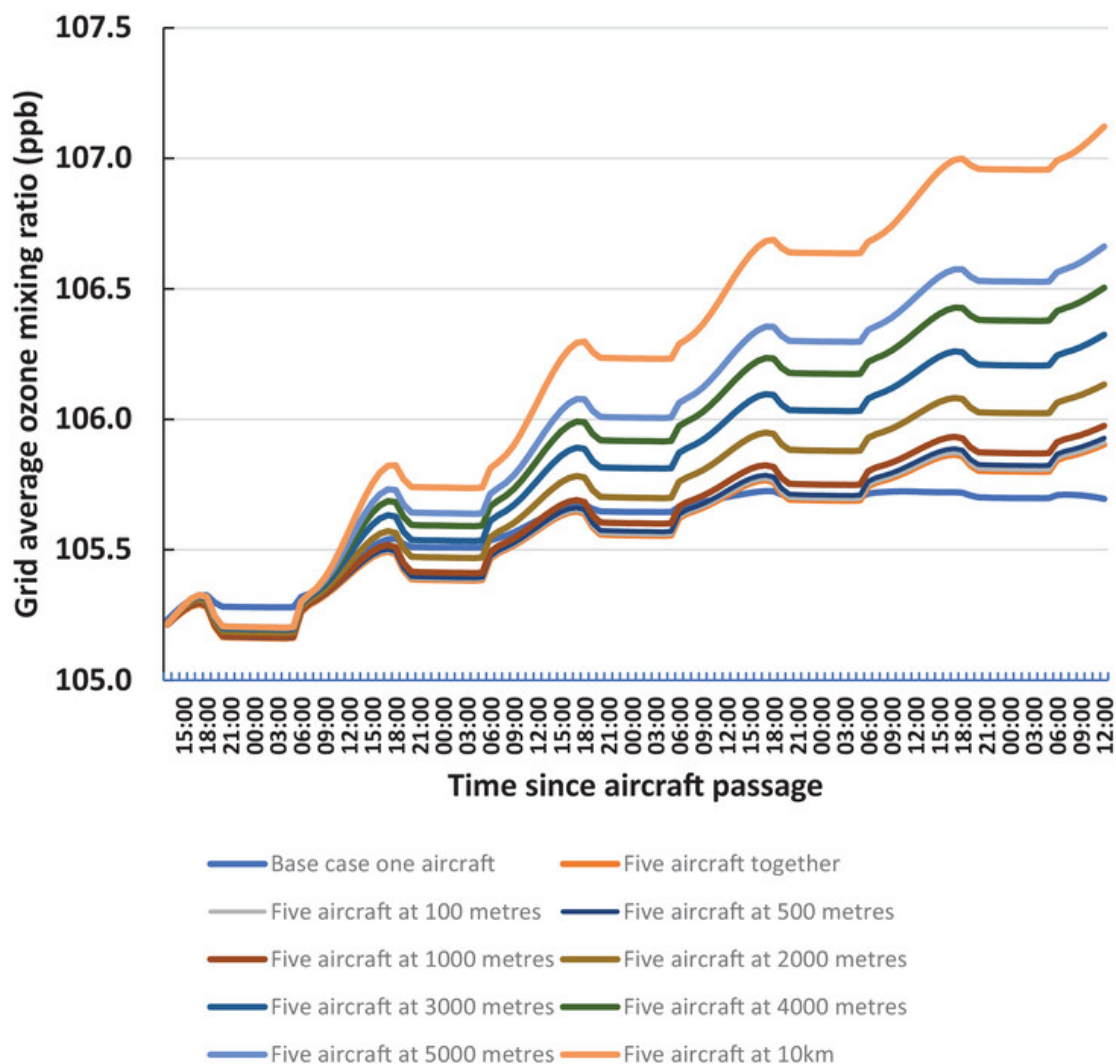


Figure 2.6: O₃ mixing ratios as a function of distance from the flight path at increasing times after the passage of an aircraft, each offset by 3 ppb.

In the next application of the aircraft plume dispersion model, the case of five aircraft flying at the same time but separated in space are considered. To begin with the five aircraft flying together without any spatial separation are reported and then further cases where they are separated by 100 m, 500 m, and so on up to 10 km are investigated.

2 Potential to Offset Aviation NO_x-induced Warming Through Formation Flight and Chemical Saturation

Figure 2.7 shows the time development over 5 days of the 99.9 km grid average O₃ for a base case with one aircraft flying, five aircraft flying together and five aircraft with spatial separations increasing from 100 m to 10 km.

In each plot, the grid average O₃ perpendicular to the flight path shows a steady increase over the 5 days. This increase has small depressions during night-time when the NO–NO₂–O₃ system adjusts to the lack of sunlight. The steady increase is driven by photochemical ozone production because the chosen initial conditions for these experiments have NO levels above the local NO_x compensation point. Initially the plots for the five aircraft lie below the one aircraft case because of the more intense O₃ depletion found with more aircraft. However, the situation reverses after 1 day and O₃ levels in the five aircraft cases then lie above the one aircraft case. After further time has elapsed, the five aircraft cases begin to diverge with the cases with wider separations diverging the greatest.

2 Potential to Offset Aviation NO_x-induced Warming Through Formation Flight and Chemical Saturation

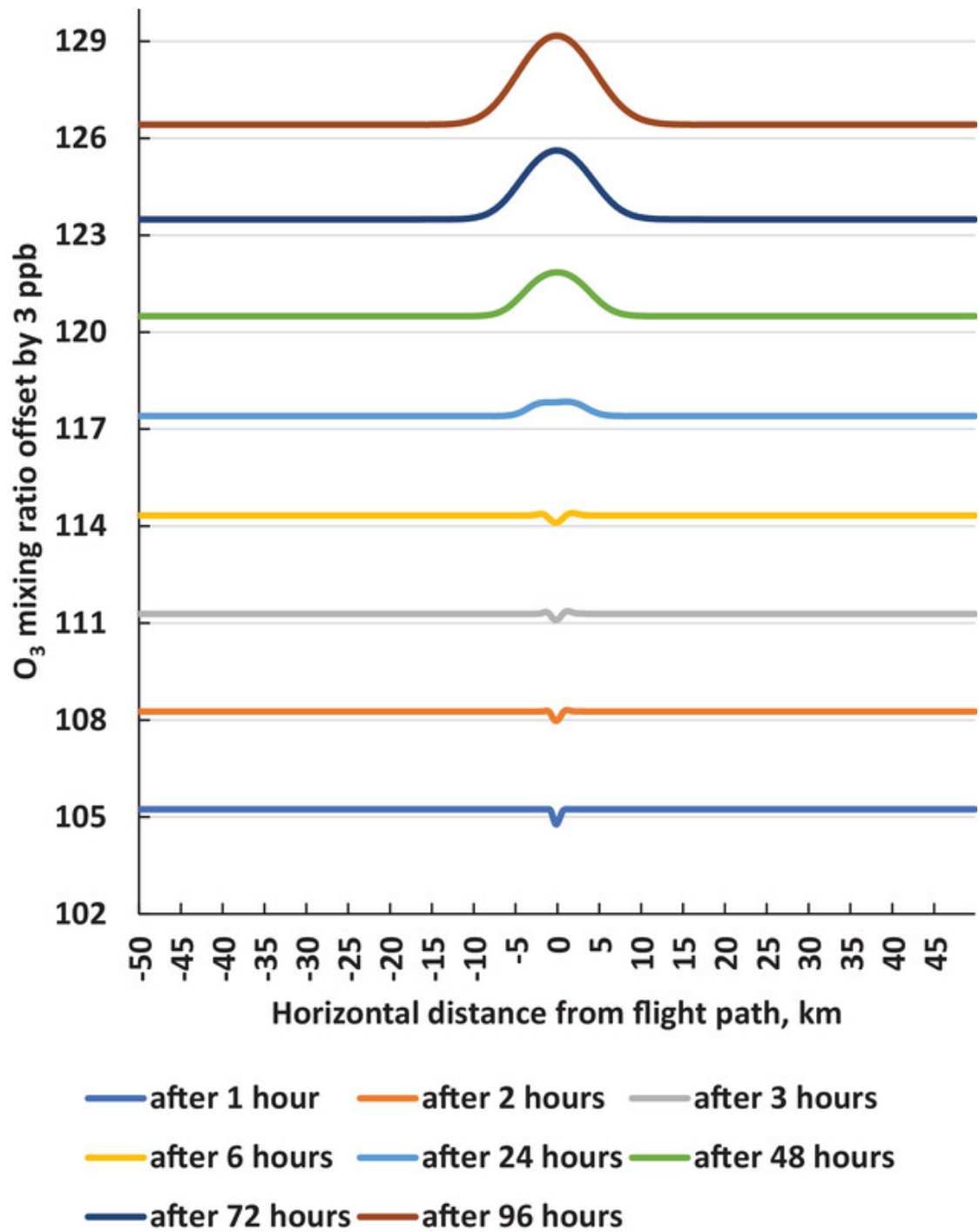


Figure 2.7: Time development of the average O₃ mixing ratios across the 99.9 km grid for a base case with one aircraft, five aircraft flying together and flying with horizontal separations increasing from 100 m to 10 km.

2 Potential to Offset Aviation NO_x-induced Warming Through Formation Flight and Chemical Saturation

After 5 days, the case with the 10 km separation has a grid averaged O₃ mixing ratio which is 1.2 ppb higher than the no-separation case, see Figure 2.7. This excess O₃ drops to just under 0.8 ppb with a horizontal separation of 5 km and to 0.2 ppb with a separation of 2 km. On this basis, the impact of aircraft emissions on spatially-averaged O₃ could be halved if the aircraft could maintain separations inside 4 km.

Clearly, this is a first illustrative example and our aircraft plume dispersion model is a highly simplified representation of the complex interplay between aircraft cruise and the local environment in an air traffic corridor. However, the results are encouraging enough for the continuation of model development, increasing spatial resolution, examining more complex scenarios, and assessing further proposals for the formation flying of aircraft.

2.4 Conclusion

When aircraft fly close to one another, their exhaust plumes can overlap, leading to the accumulation of emissions in local airspace. In the UTLS, where aircraft typically spend most of their flight, high emissions concentrations of NO_x and water vapor can lead to reductions in both ozone production and persistent contrail generation.

This paper initiated an investigation into the chemical response of the atmosphere to aircraft emissions saturation, in high-density airspace. This included a simple kinetic comparison, a coarse resolution analysis on a global scale, and a higher resolution box model study. The calculation using simple kinetic model shows that at cruise altitude, the amount of NO₂ required for parity in OH loss with loss due to reaction with CO and CH₄ is around 2 ppb. Similar calculation using the STOCHEM-CRI global model, and IAGOS measurement data shows that 0.5 ppb of [NO₂]_{equiv} may exist in northern midlatitudes. Northern midlatitudes would be most likely to benefit from formation flying to suppress ozone production. The box model results show that the impact of aircraft emissions on spatially averaged O₃ could be halved if the aircraft could maintain

2 Potential to Offset Aviation NO_x-induced Warming Through Formation Flight and Chemical Saturation

separations inside 4 km rather than 10 km.

The results generated from these simulations indicate a clear trend toward a climate beneficial outcome due to emissions saturation. In future papers, we aim to increase sophistication of the box model by integrating higher level plume modeling methods and a contrail model into the analysis. In addition to this, the aim is to integrate air traffic trajectories into the study, such that emissions from either real world or simulated flights can be modeled more accurately.

3 Development of a Novel Aircraft Plume Modelling Framework

In this chapter, a robust methodology is presented for estimating the chemical saturation effects that occur, as a result of overlapping aircraft exhaust plumes in busy airspace. This is done through explicitly modelling aircraft trajectories, simulating their flight performance to ascertain fuel burn and emissions data, then simulating plume dispersion to investigate how those emissions mix into the surrounding atmosphere. Then aggregating these data to a common Eulerian grid for multiple flights in the same region enables further chemical and microphysical modelling, to explore a range of responses to aircraft emissions in high-density airspace.

3.1 Introduction

A number of studies to date have conducted climate-chemistry simulations, to investigate the plume overlap effect [**<empty citation>**]. Through modelling of local plume dispersion and the ensuing chemical and microphysical transformations, they have found a general trend towards reducing NO_x- and contrail-related warming under increasing emissions concentrations. This is due to saturation effects which occur in high-emission airspace, as elaborated on in ???. Such studies do not however, provide far-reaching conclusions on the range of saturation effects that occur, under different formation flight

3 Development of a Novel Aircraft Plume Modelling Framework

configurations and atmospheric conditions. In chapter ??, the practicality of controlled plume intersection was investigated, through real-world air traffic data analysis. Under the "significant plume encounter (SPE)" criteria laid out, it was found that nearly 40% of all aircraft plumes in the high-density hotspots over mainland US and Europe, exhibited some degree of plume overlap. Chapter 2 approaches the problem from a different perspective, focusing primarily on the chemical saturation effects that would result from high NO_x emissions. A metric called NO₂ threshold was developed, to represent the propensity of a region for NO_x saturation in the upper troposphere. The chapter concluded with a first-order photochemical box model experiment, which went some way towards quantifying the chemical saturation effect in terms of formation flight and plume intersection. The motivation behind this research was to begin to identify suitable regions of the atmosphere to exploit chemical saturation effects resulting from formation flight and plume overlap.

Chapters 3 and 4 serve as a direct continuation of this body of research, aiming to provide further real world context through higher-fidelity aircraft plume modelling and local atmospheric chemistry analysis. The literature survey conducted throughout the duration of this PhD revealed a lack of pre-existing modelling architectures, suitable for simulating the emissions, dispersion and local chemical and physical evolution of overlapping aircraft exhaust plumes. Therefore it was deemed imperative to construct a novel framework to fit these requirements. This led to the development of the Gridded Plume Analysis Tool (GPAT). The following sections will delineate the underlying models, input parameters and datasets contained within the GPAT architecture.

3.2 GPAT Overview

The Gridded Plume Analysis Tool (GPAT) derives from work done by the University of Bristol Atmospheric Chemistry Research Group (ACRG), Professor Ulrich Schumann and

3 Development of a Novel Aircraft Plume Modelling Framework

colleagues at Deutsches Zentrum für Luft- und Raumfahrt (DLR), as well as Breakthrough Energy and their recent development of the *pycontrails* open-source repository for aviation climate impact modelling. Each group’s respective work was integral to the development and application of GPAT, as will become apparent throughout this chapter.

This tool was built out of a development version of the *pycontrails* library (v0.53.2), with the majority of the models and interlinking software being written in Python. The only exception to this being the photochemical box model (BOXM v2.0) written in FORTRAN 90, which is an adaptation of a legacy model (BOXM v1.0) ?? built in FORTRAN 77. All libraries, packages and tools used throughout the software development phase will be outlined explicitly where relevant.

The primary purpose of GPAT is to simulate a spatial domain in which to plot aircraft trajectories, estimate their emissions, model the dispersion and advection of the exhaust plumes carrying the emissions, then to compute the evolution of atmospheric chemistry and physics across the domain. The software architecture of the modelling framework is presented in figure 3.1. This portrays the input parameters, constituent models and the input and output datasets related to each model.

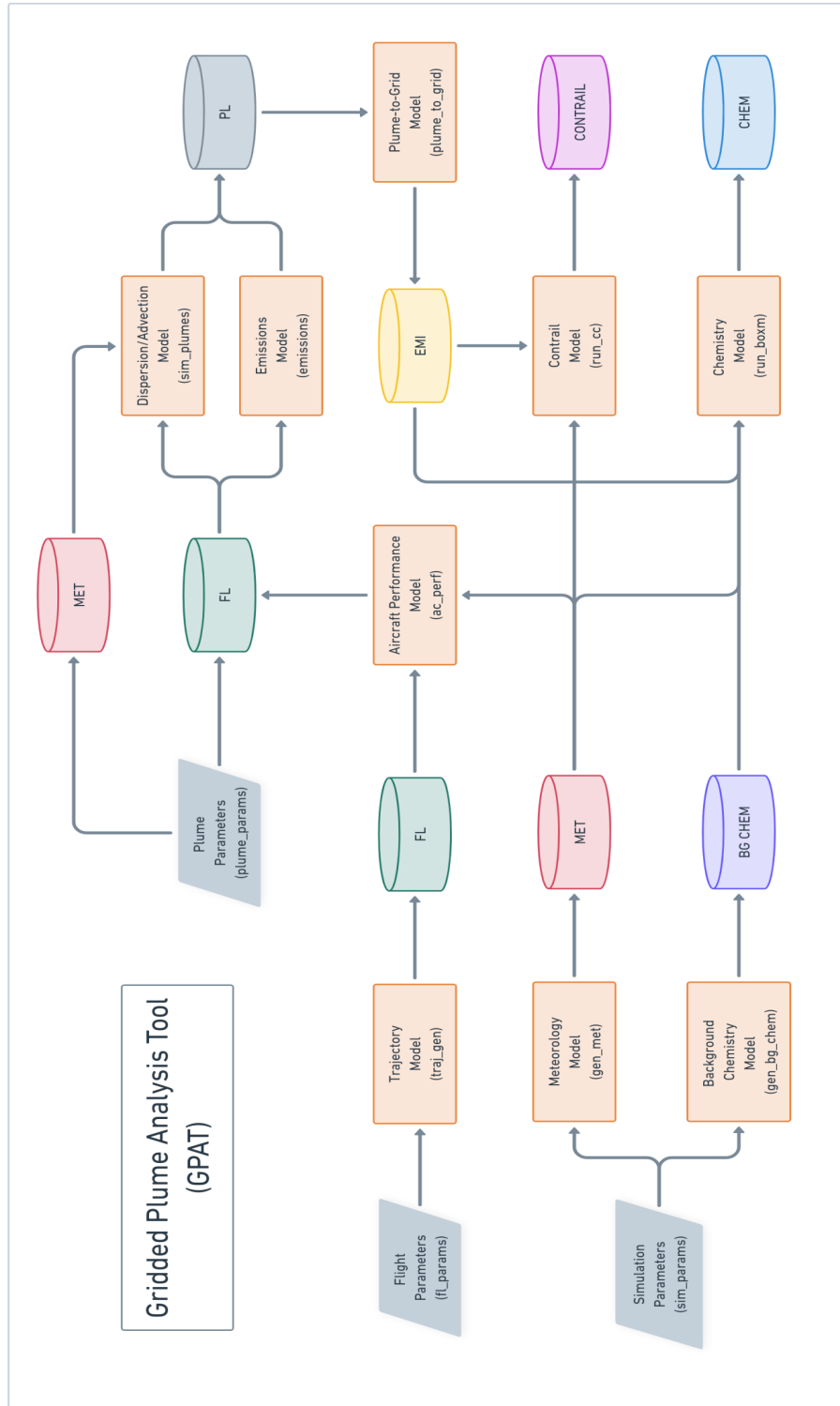


Figure 3.1: GPAT process flow schematic portraying the input parameters (rhombi), constituent models (boxes) and flow of data through a range of datasets (cylinders).

3.3 Input Parameters

The Gridded Plume Analysis Tool requires three sets of user-defined input parameters, to provide the necessary information to conduct a model run. This comprises flight parameters related to aircraft and flight properties, plume parameters pertaining to the characteristics of the associated aircraft exhaust plumes, and lastly the simulation parameters, required to instantiate the physical and chemical modelling environment.

3.3.1 Flight Parameters (`fl_params`)

The flight parameters give information on the flights taking place in the simulation. This includes the start time, duration and time step of the flight data to be generated, the aircraft types, speed, heading and initial coordinates of the first aircraft. For formation flight scenarios, it also contains the longitudinal (dx), lateral (dy) and vertical (dz) separation minima between the leader and follower aircraft, as well as the number of aircraft in the simulation. It is assumed that follower aircraft are travelling at the same speed and along the same heading. Table ?? shows these parameters as they are read by the software, along with a description of their use, units and an example value.

Parameter	Description	Example Value
t0_fl	Flight start time	2022-01-01 12:00:00
rt_fl	Flight run time [min]	60
ts_fl	Flight time step [min]	2
ac_type	Aircraft type	A320
fl0_speed	Aircraft speed [m/s]	100
fl0_heading	Aircraft heading [deg]	45
fl0_coords0	Leader aircraft initial coordinates [lat. deg, lon. deg, alt. m]	(45.0, 45.0, 12,000)
sep_dist	Separation minima [dx m, dy m, dz m]	(1000, 100, 0)
n_ac	Number of aircraft	5

3.3.2 Plume Parameters (plume_params)

These are the parameters that determine how the exhaust plumes, and emissions contained within them, disperse and advect spatially over time. This includes the integration time step, max age of each plume, initial dimensions, wind shear, horizontal and vertical resolution of the plume simulation, as well as the number of Gaussian slices for the plume aggregation step (for more on this, see section ??). Table ?? shows these parameters as they are read by the software, along with a description of their use, units and an example value.

Parameter	Description	Example Value
dt_integration	Integration time step [min]	2
max_age	Maximum age of plume segments [hours]	2
depth	Initial plume depth [m]	50
width	Initial plume width [m]	50
shear	Wind shear [1/s]	0.01
hres_pl	Horizontal resolution of plume aggregation [deg]	0.05
vres_pl	Vertical resolution of plume aggregation [m]	500
n_slices	Number of Gaussian slices to aggregate to grid	10

3.3.3 Simulation Parameters (sim_params)

This set of parameters defines the spatial and temporal domain of the simulation run, specifying the geographical bounds of the region of interest, the time frame over which the atmospheric chemistry and physics is simulated, the spatial and temporal resolution of the domain, wind effects, and species in and out of the model. Table ?? shows these parameters as they are read by the software, along with a description of their use, units and an example value.

3 Development of a Novel Aircraft Plume Modelling Framework

Parameter	Description	Example Value
t0_sim	Simulation start time	2022-01-01 12:00:00
rt_sim	Simulation run time [days]	5
ts_sim	Simulation time step [s]	20
lat_bounds	Latitude bounds [deg]	(40, 50)
lon_bounds	Longitude bounds [deg]	(40, 50)
alt_bounds	Altitude bounds [m]	(12000, 13000)
hres_sim	Horizontal resolution of simulation [deg]	500
vres_sim	Vertical resolution of simulation [deg]	10
eastward_wind	Eastward bound windspeed [m/s]	0
northward_wind	Northward bound windspeed [m/s]	0
vertical_velocity	Vertical wind velocity [m/s]	0
species_in	Emission species concentrations aggregated to sim. [ppbv]	("NO", "NO2", "CO")
species_out	Output species concentrations [ppbv]	("O3", "NO2", "NO")

3.4 Datasets

The datasets contained within the GPAT modelling framework help maintain a consistent flow of data between models, providing information about the specific flights in the run and the associated fuel burn and emissions masses throughout those flights (FL). The dispersion and advection of their corresponding plumes is captured in the Plume Dataset (PL). To investigate the ensuing climate effects of those emissions through gridded chemical and microphysical analysis, the Background Chemistry (BG_CHEM) and Meteorology (MET) Datasets is defined. The Emissions Dataset (EMI) is generated through aggregation of 1D emissions and plume geovector data to a 3D Eulerian grid.

3 Development of a Novel Aircraft Plume Modelling Framework

The Contrail (CONTRAIL) and Chemistry (CHEM) Datasets are then generated by concatenating background chemistry, meteorology and emissions datasets, followed by chemical and microphysical processing through the `run_cc` and `run_boxm` models. This final step provides insight into how the emissions present in the model affect the composition of the surrounding atmosphere, over the specified runtime of the simulation. All data structures contained within the GPAT framework are created using the Xarray Python library and pycontrails [Xarray2024; pycontrails2024]

3.4.1 Flight Dataset (FL)

The Flight Dataset is first generated when the user-defined Flight Parameters (`fl_params`) are passed through the Trajectory Model. The latitude, longitude, altitude coordinates as well as the waypoint number are calculated for each timestep throughout the flight, according to `t0_fl`, `ts_fl` and `rt_fl`. Flight attributes are also assigned to this dataset as metadata; the flight's unique identifier (UID), aircraft type, number of engines, wingspan, maximum mach number, maximum altitude and engine UID.

The Flight Dataset is passed to the Aircraft Performance Model, which estimates aircraft fuel flow using the aircraft positional data, aircraft-specific characteristics and the instantaneous meteorological situation. The downselected meteorology data is assigned prior to the performance run. Aircraft performance is then simulated and the dataset is further populated with true airspeed, thrust, thrust setting, rate of climb or descent, engine efficiency, fuel flow, fuel burn and aircraft mass at each flight waypoint. Total fuel burn is also added to the FL dataset's attributes. An example of a fully populated Flight Dataset is provided in table ??

3.4.2 Meteorology Dataset (MET)

MET provides information on the meteorological situation over all timesteps and grid cells of the simulation region. The air temperature, water vapour concentration and wind field data are taken from the UK Met Office Unified Model (UM) for the year 1998 at a resolution of $5^\circ \times 5^\circ$ [**<empty citation>**]. T

3 Development of a Novel Aircraft Plume Modelling Framework

3.4.3 Background Chemistry Dataset (BG_CHEM)

3.4.4 Plume Dataset (PL)

3.4.5 Emissions Dataset (EMI)

3.4.6 Contrail Dataset (CONTRAIL)

3.4.7 Chemistry Dataset (CHEM)

3.5 Models and Methods

3.5.1 Trajectory Model (traj_gen)

3.5.2 Meteorology Model (gen_met)

3.5.3 Background Chemistry Model (gen_bg_chem)

3.5.4 Aircraft Performance Model (ac_perf)

3.5.5 Emissions Model (emissions)

3.5.6 Dispersion/Advection Model (sim_plumes)

3.5.7 Plume-to-Grid Model (plume_to_grid)

Instantaneous Dispersion

Gaussian Plume Aggregation

3.5.8 Contrail Model (run_cc)

3.5.9 Chemistry Model (run_boxm)

3.5.10 Validation

Mass Conversation

Box Model Outputs

3.6 Conclusion

4 Plume Overlap as A Climate Solution – A Sensitivity Study

5 Conclusion

Bibliography